



## AQ2.2: Activity Questions 2- Not Graded



This assignment will not be graded and is only for practice.

**Level 1:**

1 point

Consider a system of linear equations as follows:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 &= b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 &= b_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 &= b_3 \end{aligned}$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3$$

Let the matrix representation of the above system be  $Ax = b$ , where  $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$ ,  $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ , and  $b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$ . Let  $A_{x_i}$  be the matrix obtained by replacing the  $i$ -th column of  $A$  (i.e.,  $\begin{bmatrix} a_{1i} \\ a_{2i} \\ a_{3i} \end{bmatrix}$ ) by  $b$ , for  $i = 1, 2, 3$ . If  $\det(A_{x_i}) = 0$  for  $i = 1, 2, 3$ , then which of the following is (are) true?

☐

$x = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$  must be a solution.

☐

$x = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$  is a solution if  $\det(A) \neq 0$ .

☐

If  $\det(A) = 0$ , then we cannot conclude about the solution using Cramer's rule.

☐

If  $\det(A) \neq 0$ , then  $b = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**



$x = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$  is a solution if  $\det(A) \neq 0$ .

If  $\det(A) = 0$ , then we cannot conclude about the solution using Cramer's rule.

If  $\det(A) = 0$ , then  $b = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ .

Consider a system of linear equations

$$\begin{aligned} x_1 + x_3 &= 1 \\ -x_1 + x_2 - x_3 &= 1 \\ -x_2 + x_3 &= 1 \end{aligned}$$

Let matrix representation of the above system be  $Ax = b$ , where  $A = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix}$ ,  $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ , and  $b = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ . Let  $A_{x_i}$  be the matrix obtained by

replacing the  $i$ -th column of  $A$  (i.e.,  $\begin{bmatrix} a_{1i} \\ a_{2i} \\ a_{3i} \end{bmatrix}$ ) by  $b$ , for  $i = 1, 2, 3$ .

**(Use the above information to answer questions 2 and 3)**

1 point

Choose the set of correct options

☐

$$A_{x_1} = \begin{bmatrix} 1 & 1 & 0 \\ -1 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix} Ax_1 = \begin{bmatrix} 1 & -1 & 0 & 1 & -1 & 1 \\ 1 & -1 & 0 & 1 & -1 & 1 \\ 1 & -1 & 0 & 1 & -1 & 1 \end{bmatrix}$$

☐

$$A_{x_1} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix} Ax_1 = \begin{bmatrix} 1 & 1 & 0 & 1 & -1 & 1 \\ 1 & 1 & 0 & 1 & -1 & 1 \\ 1 & 1 & 0 & 1 & -1 & 1 \end{bmatrix}$$

☐

$$A_{x_2} = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix} Ax_2 = \begin{bmatrix} 1 & -1 & 0 & 1 & -1 & 1 \\ 1 & -1 & 0 & 1 & -1 & 1 \\ 1 & -1 & 0 & 1 & -1 & 1 \end{bmatrix}$$

☐

$$A_{x_3} = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 1 \\ 0 & -1 & 1 \end{bmatrix} A x_3 = \begin{bmatrix} \phantom{0} \\ 1-1001-1111 \\ \phantom{0} \end{bmatrix}$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$A_{x_1} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix} A x_1 = \begin{bmatrix} \phantom{0} \\ 11101-11-11 \\ \phantom{0} \end{bmatrix}$$

$$A_{x_3} = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 1 \\ 0 & -1 & 1 \end{bmatrix} A x_3 = \begin{bmatrix} \phantom{0} \\ 1-1001-1111 \\ \phantom{0} \end{bmatrix}$$

*1 point*

Choose the set of correct options.

☐

$$x_1 = -2x_1 = -2.$$

☐

$$x_2 = -2x_2 = -2.$$

☐

$$x_3 = 3x_3 = 3.$$

☐ None of the above.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$x_1 = -2x_1 = -2.$$

$$x_3 = 3x_3 = 3.$$

**Level 2:**

Consider the system of linear equations  $Ax = b$ , where  $A = \begin{bmatrix} 1 & a & 0 \\ a & 1 & 2 \\ 1 & 0 & a \end{bmatrix}$ ,  $x = \begin{bmatrix} x_1 \\ 0 \\ x_3 \end{bmatrix}$

$x = \begin{bmatrix} x_1 \\ 0 \\ x_3 \end{bmatrix}$ , and  $b = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ , and the solution for  $x$  is partially known. What is the value of

$$a^2 a^2, \text{ if } a \neq 0, \sqrt{3}, -\sqrt{3} a = 0, \quad 3$$

$\sqrt{\phantom{0}}$

, - 3

$\sqrt{\phantom{0}}$

is given? [Hint: Observe that the second row of the vector  $x$  is given as 00, which implies that  $x_2$  is

known, where  $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 2

1 point

Consider the system of linear equations  $Ax = b$ , where  $A = \begin{bmatrix} 1 & 2 \\ 3a+2 & a+3 \end{bmatrix}$ ,  $x = \begin{bmatrix} x_1 \\ 0 \end{bmatrix}$ , and  $b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ , and the solution for  $x$  is partially known.

What is the value of  $3a$ , if  $\det(A) \neq 0$  is given?

**No, the answer is incorrect.**

**Score: 0**